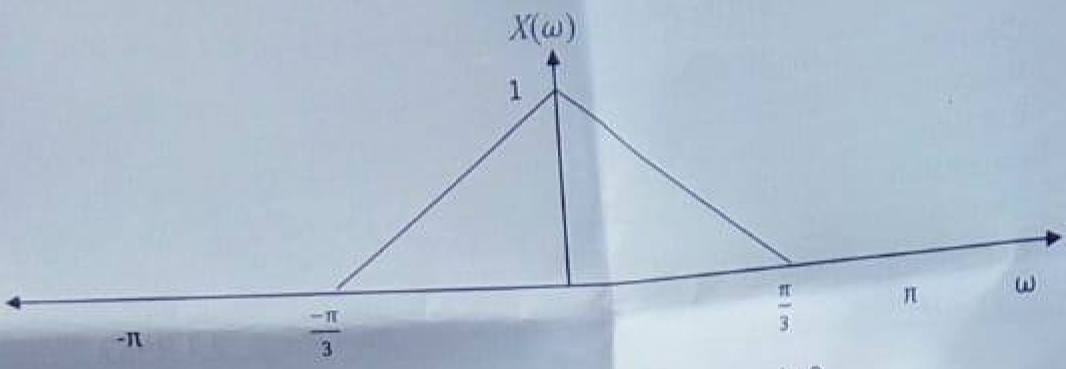

Sketch the DTFT of the sequences:

$x(\omega - \varphi)\, e^{j\omega n}$

i. $x(n)e^{-j\pi n/3}$

ii. $x(n)\cos(\frac{\pi}{6} n)$

b) Find the Discrete time Fourier transform (DTFT) of the two sided sequence:

$$x(n) = \left(\frac{1}{4}\right)^{|n|}$$

Question # 4

CLO-05 [10+25=35 Marks]

a) Consider the sequence $x(n) = \delta(n) + 2\delta(n-2) + \delta(n-3)$

i. Find the 4-point DFT of $x(n)$ $x[k] = [3, -1+i, 2, -1-i]$

ii. If $y(n)$ is the 4-point circular convolution of $x(n)$ with itself, find $y(n)$ and the 4-point DFT $Y_4(k)$

$y(n) = (5, 4, 5, 2)$

$Y_4(k) = [16, -2i, 4, 2i]$

$x(k) = \sum_{n=0}^{N-1} x(n) W_N^{k} = \sum_{0}^{3} x(n) e^{-\frac{j2\pi nk}{N}}$

b) Consider the 4-point DFT of the signal $x(n)$ ($x(n)$ is of length 4 as well) as follows.

$$X(k) = \sum_{n=0}^{3} x(n)e^{-\frac{j2\pi kn}{4}}, k = 0, ..3 \qquad (1)$$

$x[k] = [1-i, 1+9i, 1+i, -5+i]$

i. What is the total complexity of the computation of $X(k)$ using equation (1) $N = 4$

ii. Now consider the radix-2 decimation in frequency Fast Fourier transform (FFT) implementation of $X(k)$. If $x(n) = \{1, 0, -2, 3\}$, draw the corresponding data flow diagram of 4-point FFT showing each stage and calculate $X(k)$ using the FFT diagram.

$X(k) = F_1(k) + W_N^{k} F_2(k)$

$X(k+N) = F_1(k) - W_N^{k} F_2(k)$

$k = \frac{N}{2} - 1$

iii. What is the total complexity of the radix-2 decimation in time FFT algorithm? Compare it with your answer in part (a)

Question # 5

CLO-04 [7 Marks]

A sequence $x(n)$ corresponds to samples of a bandlimited signal using a sampling frequency $F_s = 10Khz$. However, the sequence should have been sampled using a sampling frequency $F_s = 12Khz$. Design a system (values of L and M) for digitally changing the sampling rate.

$\frac{L}{M} = \frac{6}{5}$

National University of Computer and Emerging Sciences, Lahore Campus

Course:	Digital Signal Processing	Course Code:	EE302
Program:	BS(EE)	Semester:	Fall 2018
Duration:	3 hours	Total Marks:	100
Paper Date:	27th Dec '18	Weight	
Section:	EE	Page(s):	2
Exam:	Final		

Student : Name: Malik Zaeem **Roll No.** 15L4452

Instruction/Notes:

- In case of more than one attempts to a question, only one will be evaluated, whichever is found first.
- Answer all the questions on the answer sheet
- Calculators are allowed

Question # 1

CLO-07 [5+5+5+3=18 Marks]

Consider the filter

$$y(n) = 0.9y(n-1) + bx(n)$$

i. Determine b such that the frequency response of the filter satisfies the condition: $|H(0)| = 1$
Hint: Find $H(\omega)$ where $H(\omega) = \frac{Y(\omega)}{X(\omega)}$ (Use Fourier transform properties)

ii. Draw pole zero plot of the system function $H(z)$ $X(\omega) = \sum x(n) e^{-j\omega n}$

iii. Determine the frequency at which $|H(\omega)| = \frac{1}{\sqrt{2}}$

iv. Would you classify this filter as low pass, high pass, or bandpass? Is this filter stable? Justify your answer.

Question # 2

CLO-06 [10+10=20 Marks]

a) Find all possible $x(n)$ associated with the Z-transform:

$$X(z) = \frac{5z^{-1}}{(3 - z^{-1})(1 - 2z^{-1})}$$

$|z| > a$

b) Find $X(z)$ and the region of convergence of of each of the following sequence: a^n

i. $x(n) = 2^n u(-n)$

ii. $x(n) = u(n + 10) - u(n + 5)$ $\frac{1}{1-A}$ $\frac{1}{1-az^{-1}}$

Question # 3

CLO-03 [10+10=20 Marks]

a) A sequence $x(n)$ has zero phase $\boxed{\text{DTFT } X(\omega)}$ as sketched in figure below.

? $e^{-j\omega n}$

Formula =